A Unified Graph-Mamba Framework for Cooperative Autonomous Driving in Mixed Traffic

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***Abstract*—Safe cooperative driving in mixed traffic is challenged by complex spatiotemporal interactions and unpredictable human behaviors. To address this, we introduce a Graph-Mamba Network (GMN) within a Multi-Agent Reinforcement Learning (MARL) framework. GMN integrates Graph Attention Networks (GATs) to capture spatial dependencies and the Mamba model to encode long-term temporal dynamics. Notably, Mamba's linear computational complexity ensures superior efficiency over Transformer-based methods for long-horizon tasks. Adopting the Centralized Training with Decentralized Execution (CTDE) paradigm, our approach balances global coordination with local decision-making. Experiments on MetaDrive demonstrate that GMN outperforms baselines in safety and robustness, advancing MARL applications in intelligent transportation systems.**

***Index Terms*—mixed Traffic, multi-agent reinforcement learning, safe cooperative driving, graph-mamba network.**

I. Introduction

Autonomous driving has advanced significantly, with learning-based policies often outperforming conventional controls [1] [2] [3]. However, mixed-traffic scenarios remain challenging due to the complex spatiotemporal interdependencies introduced by unpredictable human-driven vehicles [4] [5]. While Multi-agent reinforcement learning offers a promising alternative where traditional controllers struggle [6], existing frameworks face three key limitations: (1) inadequate modeling of explicit spatial relationships; (2) difficulty in capturing long-term dependencies under partial observability; and (3) scalability issues that hinder convergence in dense environments [7].

To address these, we propose the Graph-Mamba Network (GMN), integrating Graph Attention Networks (GAT) [8] and Mamba [9] [10] into a MARL framework. GMN synergizes GAT's dynamic interaction graphs for spatial encoding with Mamba's selective state-space models (SSMs) for efficient, linear-complexity temporal modeling. This architecture enables robust decision-making by jointly reasoning over immediate spatial contexts and long-term historical actions. The main contributions are:

- We propose GMN, a novel architecture that fuses GAT and Mamba within a MARL framework. It effectively models dynamic spatial interactions via graph structures and captures long-term temporal dependencies through efficient SSMs.
- We evaluate the proposed framework on the MetaDrive platform across challenging scenarios [11]. The results demonstrate that GMN outperforms state-of-the-art MARL baselines in terms of safety, robustness, and scalability, validating its effectiveness in mixed-traffic environments.

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II. Related Work

A. Driving Decision-Making in Mixed-Traffic

Context-aware decision-making is crucial for safety and efficiency in the complex dynamics of mixed traffic [12]. While game-theoretic and cooperative strategies have improved performance in specific scenarios like ramp merging and congestion mitigation [13]–[15], they often suffer from poor scalability and insufficient long-term temporal modeling in highly dynamic environments. MARL offers a decentralized alternative for collaborative driving [16], with recent works utilizing Graph Neural Networks (GNNs) and attention mechanisms to structure agent interactions [17], [18]. However, existing MARL frameworks still face challenges in exploration efficiency and capturing long-term dependencies, limiting their applicability in large-scale mixed-traffic scenarios.

B. Enhancing MARL with GNN and Mamba

To address spatial complexity, GNNs have been integrated into MARL to explicitly model topological agent interactions, facilitating effective reasoning in dense traffic [19], [20]. However, GNNs alone struggle to capture the long-term temporal dependencies essential for understanding delayed rewards. The Mamba architecture addresses this by offering efficient state-space sequence modeling with linear computational complexity, making it ideal for scalable multi-agent systems [21]. By integrating Mamba with GNNs, recent studies have constructed robust spatiotemporal backbones that significantly enhance long-term reasoning and temporal consistency in cooperative driving tasks [22].

III. Method

A. Problem Definition

We formalize the cooperative driving decision-making problem in mixed traffic as a Decentralized Partially Observable Markov Decision Process (DEC-POMDP). Specifically, a DEC-POMDP is defined by the tuple $\langle N, S, \{A_i\}, \{O_i\}, \{R_i\}, T, \Omega, \gamma \rangle$, where N denotes a finite set of agents involved in the cooperative driving task, typically representing autonomous vehicles in the mixed-traffic setting. The global state space S represents the complete environmental configuration, which is not fully observable to any single agent. Each agent i independently selects actions from its own action space A_i based on partial observations O_i obtained through local perception. The environment changes according to a stochastic transition function $T(s'|s,a)$, which defines the probability of transitioning to a new state s' given the current state s and joint action $a \in \prod_i A_i$. Observations are generated by the function $\Omega(o|s,a,s')$, where $o = (o^1, ..., o^N)$ denotes the joint observation vector after the state transition. The reward function $R_i(s,a,s')$ specifies the immediate feedback for agent i, and $\gamma \in [0,1)$ is a discount factor that controls the relative importance of future versus immediate rewards.

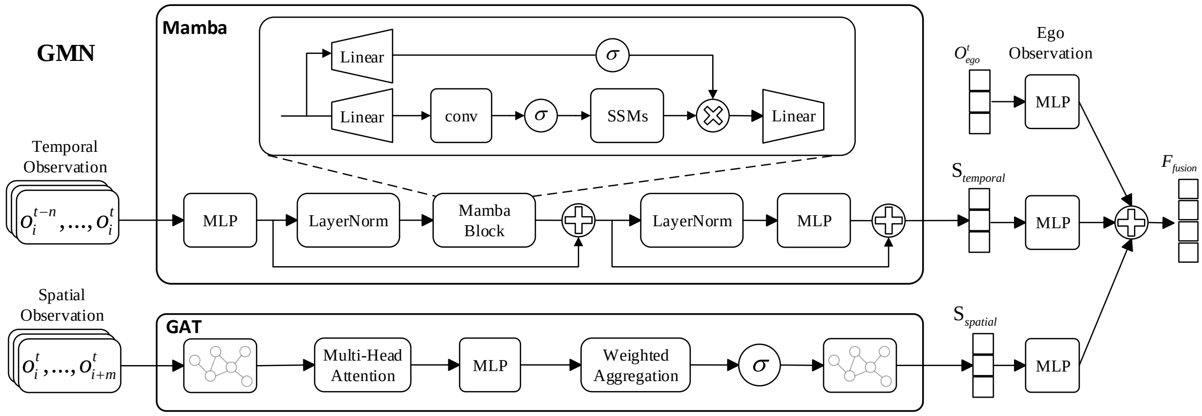

Fig. 1. Overall framework of GMN-MAPPO. The model processes temporal, spatial, and ego observations separately through dedicated MLP layers. Temporal dependencies are captured by a Mamba block, while spatial interactions are modeled via a GAT module. Ego features are processed independently. Outputs are fused and passed through additional MLP layers to produce final representations for policy optimization. σ denotes non-linear activation functions (e.g., ReLU).

B. Algorithm Framework

We propose a novel Graph Mamba Network-based MARL framework for mixed-traffic scenarios. As shown in Fig. 1, the architecture fuses spatial features from Graph Attention Networks and temporal features from Mamba via element-wise addition. These enriched representations enhance the centralized critic within a MAPPO structure, enabling stable training with decentralized execution.

1) Spatial Modeling via Graph Attention Network: In the GMN framework, spatial interactions are modeled as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes $v_i \in \mathcal{V}$ represent vehicles. Edges (v_i, v_j) are established between vehicles within a 50-meter radius. This threshold aligns with reliable onboard sensor ranges [23] and guarantees a time-to-collision buffer of over 3 seconds at typical urban speeds. Consequently, it safely bounds the interactive zone, ensuring sufficient proactive decision time while filtering out distant vehicles to prevent attention noise. Each node is associated with a spatial feature vector $\mathbf{s}_i \in \mathbb{R}^d$, including vehicle states and road-line information. The full feature matrix is denoted by $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_N]$, with each $\mathbf{s}_i$ defined as:

$$\mathbf{s}_i = \left[\text{Pos}_i,\ d_{\text{proj}}^{\theta},\ d_{\text{proj}}^{\perp},\ r_{\text{lane}},\ \theta_{\text{lane}},\ \text{Direction}_{\text{lane}}\right] \tag{1}$$

where Pos_i represents the position of the i-th vehicle in the local coordinate system, d_{proj}^{θ} and $d_{\text{proj}}^{\perp}$ are the longitudinal and lateral projections to a route checkpoint, r_{lane} is the lane curvature radius, θ_{lane} denotes the lane angle, and $\text{Direction}_{\text{lane}}$ indicates lane orientation.

These features are linearly transformed via $\mathbf{x}_i = W\mathbf{s}_i$, $W \in \mathbb{R}^{F' \times d}$ is a learnable projection matrix. GAT then computes pairwise attention coefficients to weigh neighboring node influences:

$$\mathbf{h}_i' = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W\mathbf{s}_j\right) \tag{2}$$

with the normalized attention weights given by:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}(a^\top[W\mathbf{s}_i \| W\mathbf{s}_j])\right)}{\sum_{k \in \mathcal{N}(i)} \exp\left(\text{LeakyReLU}(a^\top[W\mathbf{s}_i \| W\mathbf{s}_k])\right)} \tag{3}$$

The node-level embeddings $\mathbf{H}' = [\mathbf{h}_1', \mathbf{h}_2', \ldots, \mathbf{h}_N']$ encode both ego and relational spatial information. To derive a global permutation-invariant spatial summary, we apply mean pooling:

$$\mathbf{s}_{\text{spatial}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{h}_i' \tag{4}$$

where $\mathbf{s}_{\text{spatial}} \in \mathbb{R}^{F'}$ serves as a compact representation of the spatial configuration.

2) Temporal Modeling via Mamba: To model long-term temporal dependencies for AVs, we adopt Mamba, a structured state-space model optimized for sequence modeling. Unlike recurrent or attention-based models, Mamba leverages a discretized linear dynamical system with gating mechanism, achieving linear complexity in both time and memory.

The underlying discrete-time state-space system is defined as:

$$\mathbf{h}_{t+1} = \mathbf{A}\mathbf{h}_t + \mathbf{B}\mathbf{u}_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}\mathbf{u}_t \tag{5}$$

where $\mathbf{u}_t$ is the input at time t, $\mathbf{h}_t$ is the latent state, and A, B, C, D are learnable parameters. To enhance temporal selectivity, Mamba applies a gating mechanism to modulate state transitions:

$$\tilde{\mathbf{h}}_t = \text{Gate}(\mathbf{u}_t) \odot \mathbf{h}_t \tag{6}$$

This formulation enables Mamba to efficiently capture temporal patterns across long horizons while maintaining stable gradients and runtime efficiency.

The input to Mamba consists of a sequence of ego vehicle states over the past n steps: $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_n]$, where each $\mathbf{t}_i$ is defined as:

$$\mathbf{t}_i = [\text{Dist}_{\text{L}}, \text{Dist}_{\text{R}}, \Delta\theta, v, \delta, \text{Throttle}, \text{Brake}, \delta_{\text{last}}\omega, y] \tag{7}$$

with features including distances to lane boundaries, heading deviation, velocity, steering angles, control inputs, yaw rate, and lateral position. These temporal features are fed into Mamba to yield a compact temporal encoding:

$$\mathbf{s}_{\text{temporal}} = \text{Mamba}(\mathbf{T}) \tag{8}$$

The spatial encoding $\mathbf{s}_{\text{spatial}}$ and temporal encoding $\mathbf{s}_{\text{temporal}}$ are dimensionally aligned via a multi-layer perceptron (MLP), and then fused via element-wise addition:

$$\mathbf{z} = \mathbf{s}_{\text{spatial}} + \mathbf{s}_{\text{temporal}} \tag{9}$$

forming a comprehensive spatiotemporal feature vector that supports context-aware and trajectory-consistent policy learning.

C. Simulation Environment

1) Mixed Traffic Setup: To emulate real-world conditions, the simulation features mixed traffic comprising AVs controlled by our proposed RL policy and HVs governed by the Intelligent Driver Model (IDM). HVs exhibit varying behaviors (conservative, normal, aggressive) to induce stochastic interactions, enabling robust policy evaluation across dynamic scenarios.

2) Observation Space: The observation $\mathbf{o}_i \in \mathbb{R}^{91}$ comprises three normalized components: $\mathbf{o}_i = [\mathbf{o}_{\text{ego}}, \mathbf{o}_{\text{navi}}, \mathbf{o}_{\text{cloud}}]$. Specifically, $\mathbf{o}_{\text{ego}} \in \mathbb{R}^9$ encodes vehicle dynamics (e.g., velocity, steering) and lane alignment; $\mathbf{o}_{\text{navi}} \in \mathbb{R}^{10}$ provides navigation cues such as checkpoint distances and curvature; and $\mathbf{o}_{\text{cloud}} \in \mathbb{R}^{72}$ captures LiDAR-like spatial features of the surroundings. All features are scaled to $[0, 1]$ for stable training.

3) Action Space: The policy outputs a continuous action vector $a_i = [a_{\text{steer}}, a_{\text{throttle}}] \in [-1, 1]^2$. a_{steer} regulates lateral steering, while a_{throttle} governs longitudinal dynamics, with positive values for acceleration and negative for braking. This formulation enables smooth, high-frequency control in mixed traffic.

4) Reward Function: The reward function consists of both dense and sparse components, designed to guide AVs toward safe, efficient, and goal-directed behaviors in mixed-traffic scenarios. The overall reward for each autonomous vehicle is defined as:

$$R = R_{\text{driving}} + R_{\text{speed}} + R_{\text{lateral}} + R_{\text{termination}},$$

where each component reflects a distinct aspect of reasonable driving behavior. R_{driving} encourages continuous forward progress along the intended route by measuring the longitudinal distance traveled within the lane. R_{speed} promotes time-efficient driving by rewarding higher velocities within safe operational limits. R_{lateral} penalizes deviations from the lane centerline, thereby enhancing lateral stability and adherence to road geometry. Finally, $R_{\text{termination}}$ provides sparse, outcome-based feedback at the end of each episode: a success reward is granted when the agent reaches its destination, while penalties are applied for off-road excursions, collisions with other vehicles, or impacts with static obstacles. This combination effectively balances safety, efficiency, and task completion in the reward function.

IV. Experiments

We evaluate GMN-MAPPO in MetaDrive using a three-lane mixed-traffic roundabout with 10 RL and 10 IDM agents. Performance is assessed via benchmarks, ablation studies, and generalization tests across diverse driving styles, densities, and maps.

A. Comparison with Baseline Algorithms

We compare the performance of our proposed GMN-MAPPO framework with several baselines, including MAPPO, CoPO, and GT-MAPPO, in the roundabout mixed-traffic scenario. Specifically, GT-MAPPO (Graph Transformer MAPPO) mirrors GMN-MAPPO by using a GAT for spatial encoding, but replaces Mamba with a standard Transformer for temporal modeling. The training curves for success rate, episode reward, crash rate, and out-of-lane rate are presented in Fig. 2, and the final quantitative results are summarized in Table I.

Overall, GMN-MAPPO achieves superior performance across all evaluation metrics, recording the highest success rate (83.21%), the highest average reward (253.50), the lowest crash rate (14.40%), and the lowest out-of-lane rate (2.36%). In comparison, MAPPO simply concatenates local and neighboring information without structured

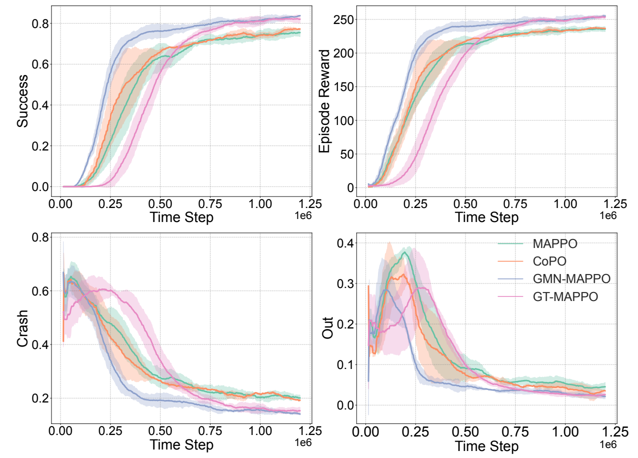

Fig. 2. The training curves of the baseline algorithms and our GMN-MAPPO in the roundabout scenario, including success rate, reward, crash rate and out rate.

TABLE I
SUMMARY OF TRAINING RESULTS IN THE ROUNDABOUT SCENARIO

Algorithm	Success(%)	Reward	Crash(%)	Out(%)
MAPPO	75.21 ± 1.51	235.62 ± 2.79	20.52 ± 0.97	4.37 ± 0.67
CoPO	77.11 ± 1.19	237.06 ± 0.65	19.75 ± 0.45	2.37 ± 0.61
GT_MAPPO	82.17 ± 1.28	253.39 ± 1.73	15.33 ± 0.93	2.43 ± 0.39
GMN_MAPPO	$\mathbf{83.21} \pm 1.24$	$\mathbf{253.50} \pm 2.20$	$\mathbf{14.40} \pm 0.34$	$\mathbf{2.36} \pm 0.61$

spatial-temporal modeling, resulting in a lower success rate (75.21%) and a higher crash rate (20.52%). CoPO enhances the reward structure by incorporating cooperative incentives, improving the success rate to 77.11% and reducing the crash rate to 19.75%. However, these improvements remain limited when compared to GMN-MAPPO.

GT-MAPPO outperforms both MAPPO and CoPO, achieving a success rate of 82.17% and a crash rate of 15.33%. However, GMN-MAPPO exhibits significantly better training efficiency, with training curves rising more rapidly in the early stages, indicating faster convergence. This enhancement is attributed to the higher computational efficiency and more effective long-horizon dependency modeling provided by Mamba, in contrast to standard Transformer architectures. While standard Transformers have quadratic complexity in their self-attention operations relative to sequence length, Mamba operates with linear complexity, making it adept at modeling long-term sequences. As a result, GMN-MAPPO facilitates faster training and improved overall performance.

B. Ablation Studies

In this section, we investigate the contributions of the spatial and temporal modules within GMN-MAPPO by introducing four ablative variants. Specifically, GT-MAPPO replaces Mamba with a Transformer, M-MAPPO and T-MAPPO remove the GAT encoder while retaining Mamba or Transformer, and G-MAPPO eliminates temporal modeling entirely.

The training curves and final success rates are presented in Fig. 3. GMN-MAPPO achieves the highest success rate (83.21%) and the fastest convergence, demonstrating the advantages of jointly modeling spatial interactions and temporal dependencies with GAT and Mamba. GT-MAPPO shows a slightly lower success rate (82.17%) and converges more slowly, indicating that Mamba provides greater efficiency for long-horizon modeling. M-MAPPO and T-MAPPO

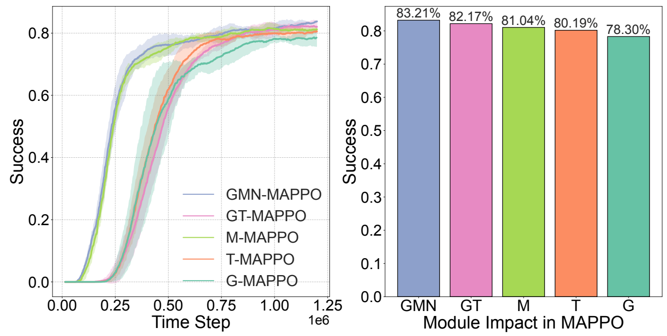

Fig. 3. Ablation study results. (a) Training curves of different model variants. (b) Final success rates comparison. GMN-MAPPO achieves the fastest convergence and highest final performance, highlighting the importance of structured spatial and efficient temporal modeling.

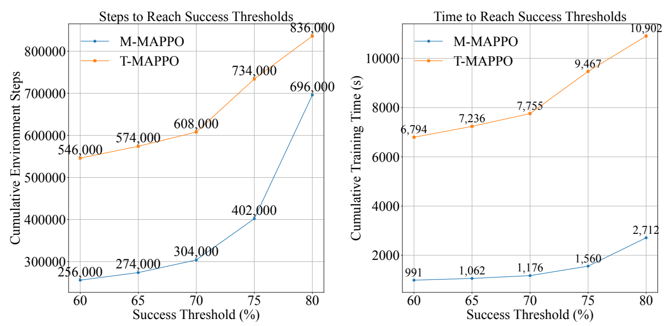

Fig. 4. Comparison of cumulative environment steps against time for M-MAPPO and T-MAPPO across different success thresholds. M-MAPPO needs fewer steps and less training time.

further decrease to success rates of 81.04% and 80.19%, respectively, underscoring the critical importance of capturing inter-agent spatial relationships. G-MAPPO records the lowest success rate (78.30%), confirming the necessity of long-term temporal modeling. Overall, these results highlight that both spatial and temporal modeling modules are essential for effective decision-making in dynamic mixed-traffic scenarios.

To further investigate the impact of temporal modeling efficiency, we analyze the cumulative environment steps and training time required for M-MAPPO and T-MAPPO in relation to success. As shown in Fig. 4, M-MAPPO requires fewer steps and less training time to achieve the same success. Specifically, to reach an 80% success rate, M-MAPPO takes only 2,712 seconds, while T-MAPPO requires 10,902 seconds, resulting in a gap of approximately 8,190 seconds. Additionally, M-MAPPO achieves comparable success rates with significantly fewer cumulative environment steps, reflecting its higher efficiency in extracting meaningful long-term temporal features. This substantial difference highlights the superior efficiency of Mamba's linear-time sequence modeling for long-horizon, dynamic decision-making tasks. In summary, the ablation studies demonstrate that GMN-MAPPO achieves both superior training efficiency and higher final performance compared to all baselines in mixed-traffic scenarios.

C. Zero-Shot Generalization Performance

To evaluate the generalization and robustness of our method, we conduct zero-shot tests across various HV driving styles, traffic densities, and traffic maps. All policies are trained in a three-lane roundabout with 10 HVs and 10 AVs (base scenario), and they

TABLE II
ZERO-SHOT GENERALIZATION PERFORMANCE: SUCCESS RATE (%).

Scenario	GMN-MAPPO	GT-MAPPO	CoPO	MAPPO
Base scenario	$\mathbf{84.35} \pm 2.07$	82.55 ± 1.57	77.53 ± 1.87	75.80 ± 2.04
HV Driving Styles				
Conservative	$\mathbf{88.19} \pm 2.11$	85.81 ± 1.24	81.14 ± 1.48	77.82 ± 1.92
Aggressive	$\mathbf{81.50} \pm 3.05$	80.57 ± 2.53	73.30 ± 3.17	71.18 ± 3.15
Mixed	$\mathbf{82.42} \pm 1.19$	81.28 ± 1.97	76.06 ± 2.15	74.08 ± 1.74
Traffic Densities				
10 HVs & 15 AVs	$\mathbf{79.68} \pm 3.34$	78.66 ± 2.06	74.91 ± 2.12	71.08 ± 2.25
15 HVs & 10 AVs	$\mathbf{75.96} \pm 2.45$	75.75 ± 2.35	67.84 ± 2.82	63.60 ± 2.84
15 HVs & 15 AVs	$\mathbf{70.28} \pm 3.47$	69.23 ± 3.45	60.62 ± 3.06	56.78 ± 3.42
Traffic Maps				
4-lane Rdbt	$\mathbf{90.19} \pm 3.09$	88.89 ± 1.59	86.12 ± 1.75	81.09 ± 2.35
3-lane Intsc	$\mathbf{70.47} \pm 4.16$	69.96 ± 3.82	59.91 ± 2.65	57.21 ± 3.38
4-lane Intsc	$\mathbf{76.69} \pm 2.78$	76.25 ± 2.07	72.95 ± 2.55	67.98 ± 3.03

are directly tested on unseen scenarios without any fine-tuning. The evaluation results are presented in Table II. Notably, we consider four HV driving styles: normal, aggressive, conservative, and mixed. The mixed driving style randomly adopts one of these styles with equal probability, reflecting the diverse behaviors of real-world human drivers.

Overall, GMN-MAPPO consistently achieves the best zero-shot generalization performance across all testing scenarios. In the base scenario, GMN-MAPPO attains an 84.35% success rate, outperforming GT-MAPPO (82.55%), CoPO (77.53%), and MAPPO (75.80%). When evaluating under different HV driving styles, GMN-MAPPO maintains strong robustness, achieving 88.19% success in conservative settings and 81.50% under aggressive conditions. GMN-MAPPO maintains a consistently higher success rate compared to GT-MAPPO. In contrast, MAPPO exhibits notable degradation under aggressive driving, with its success rate dropping to 71.18%. Regarding variations in traffic density, GMN-MAPPO exhibits better scalability, with its success rate declining more gradually as the number of vehicles increases. Specifically, under the most congested condition (15 HVs and 15 AVs), GMN-MAPPO achieves 70.28%, while GT-MAPPO achieves 69.23%, both significantly outperforming CoPO (60.62%) and MAPPO (56.78%). Across different traffic maps, GMN-MAPPO again achieves the highest success rates, reaching 90.19% on the four-lane roundabout, 70.47% on the three-lane intersection, and 76.69% on the four-lane intersection.

These results demonstrate that GMN-MAPPO consistently achieves the best zero-shot generalization performance across various HV driving styles, traffic densities, and traffic maps. By effectively integrating GAT-based spatial interaction modeling with Mamba-based long-term temporal sequence encoding, GMN-MAPPO can more accurately model high-dynamic multi-agent interactions and make reliable cooperative driving decisions in mixed traffic. The experiment video can be found here: https://drive.google.com/file/d/15irjhPzC4OvMZmBdBnViMQijDs21nR0Q/view?usp=sharing.

V. CONCLUSION

This paper investigates cooperative autonomous driving in mixed-traffic scenarios using multi-agent reinforcement learning. We propose GMN-MAPPO, a MARL framework for mixed-traffic driving that integrates Graph Attention Networks and Mamba to capture dynamic spatial interactions and long-term temporal dependencies. The approach demonstrates strong scalability and generalization across diverse traffic conditions. Future work will explore LLM-guided strategies to further address scaling challenges in complex environments.

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